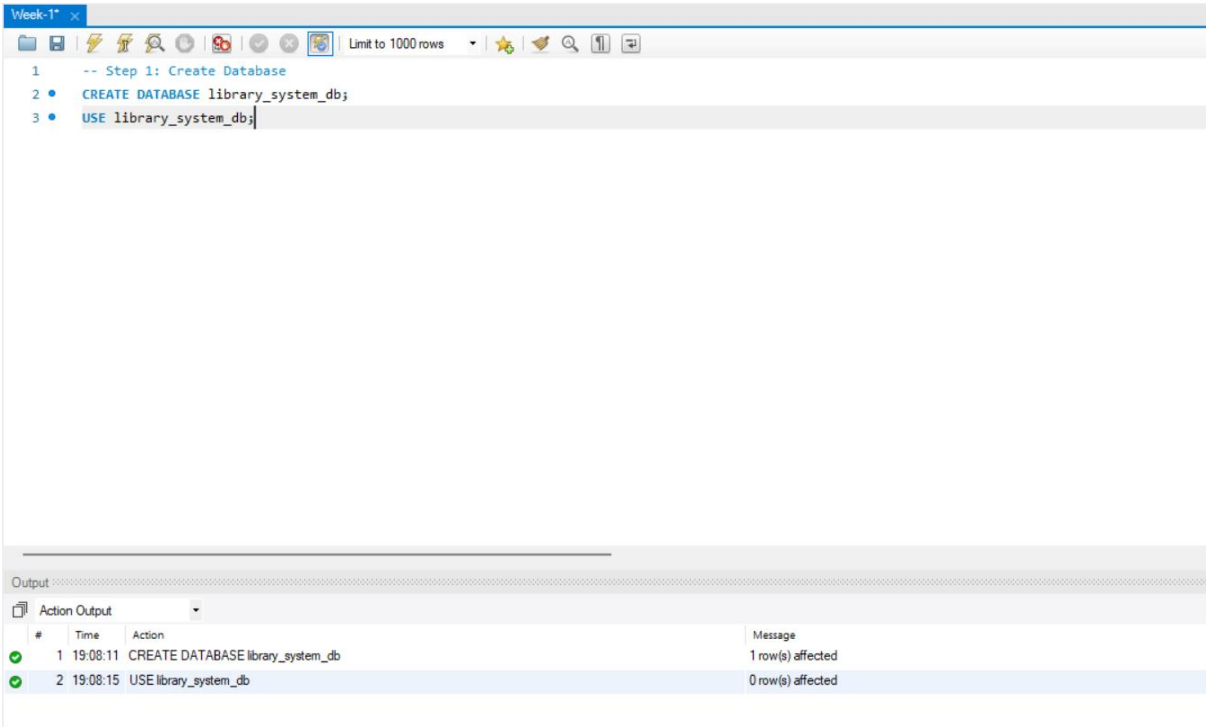
Week - 1

Ili.Harshika (2520030607)

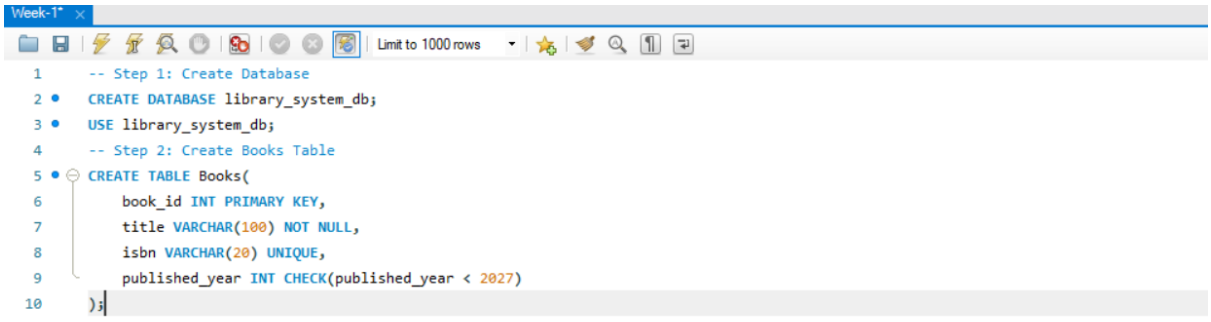
Question :

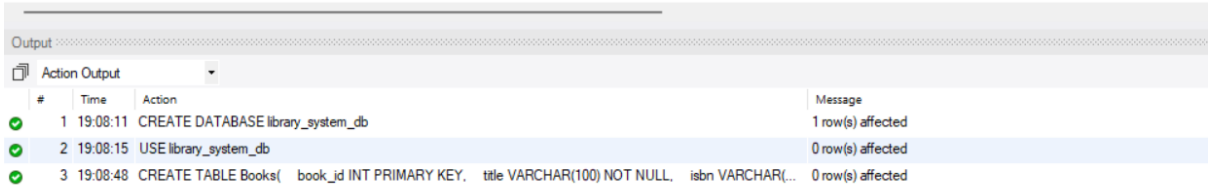
A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called **BookFlow**. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named **bookflow_db**. Then, design two tables: a **Books** table containing the columns **book_id**, **title**, **isbn**, and **published_year**, and a **Members** table containing the columns **member_id**, **full_name**, and **email**. Ensure data integrity by setting **book_id** and **member_id** as **PRIMARY KEYs**, applying a **NOT NULL** constraint to the **title** column so that every book must have a name, enforcing **UNIQUE** constraints on both **isbn** and **email** to prevent duplicate entries, and adding a **CHECK** constraint to ensure that the **published_year** is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the **Books** and **Members** tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as **CREATE DATABASE**, **CREATE TABLE**, and **INSERT** statements.

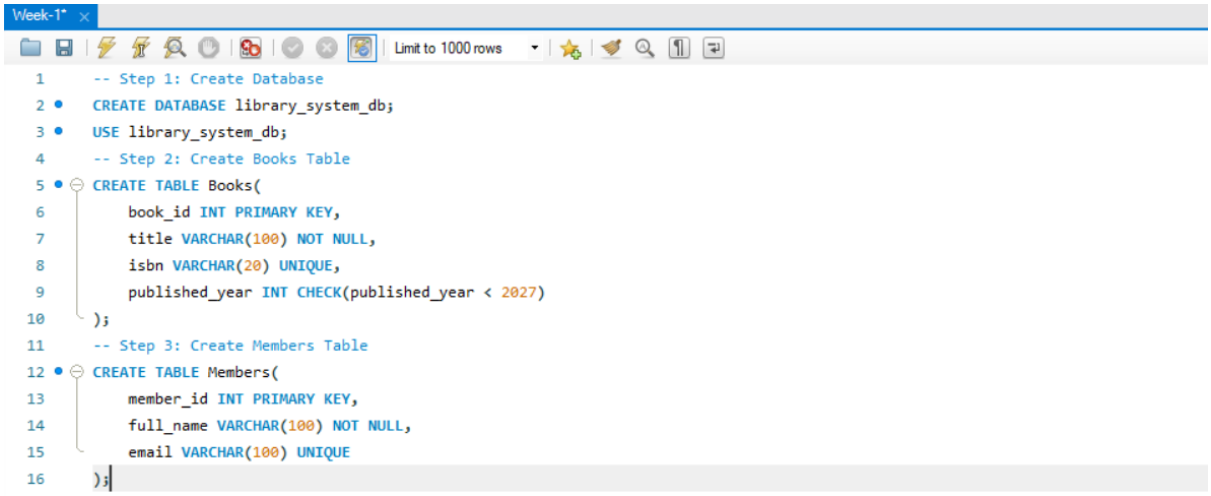
Step 1 — Create Database

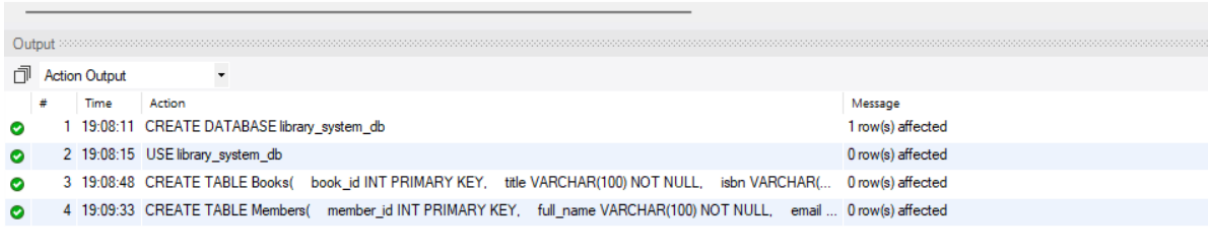


Step 2 — Create Books Table (DDL + Constraints)









Week-1*

Limit to 1000 rows

```
11    -- Step 3: Create Members Table
12    CREATE TABLE Members(
13        member_id INT PRIMARY KEY,
14        full_name VARCHAR(100) NOT NULL,
15        email VARCHAR(100) UNIQUE
16    );
17    -- Step 4: Describe Books Table
18    DESCRIBE Books;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	
title	varchar(100)	NO		NULL	
isbn	varchar(20)	YES	UNI	NULL	
published_year	int	YES		NULL	

Result 1

Output

Action Output

#	Time	Action	Message
1	19:08:11	CREATE DATABASE library_system_db	1 row(s) affected
2	19:08:15	USE library_system_db	0 row(s) affected
3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected
4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, emai...	0 row(s) affected
5	19:10:02	DESCRIBE Books	4 row(s) returned

Week-1

Limit to 1000 rows

```
13        member_id INT PRIMARY KEY,
14        full_name VARCHAR(100) NOT NULL,
15        email VARCHAR(100) UNIQUE
16    );
17    -- Step 4: Describe Books Table
18    DESCRIBE Books;
19    -- Step 5: Describe Members Table
20    DESCRIBE Members;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	
full_name	varchar(100)	NO		NULL	
email	varchar(100)	YES	UNI	NULL	

Result 2

Output

Action Output

#	Time	Action	Message
1	19:08:11	CREATE DATABASE library_system_db	1 row(s) affected
2	19:08:15	USE library_system_db	0 row(s) affected
3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected
4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, emai...	0 row(s) affected
5	19:10:02	DESCRIBE Books	4 row(s) returned
6	19:10:39	DESCRIBE Members	3 row(s) returned

Week-1*

```
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    isbn VARCHAR(20) UNIQUE,
    published_year INT CHECK(published_year < 2027)
);
-- Step 3: Create Members Table
CREATE TABLE Members(
    member_id INT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE
);
-- Step 4: Describe Books Table
DESCRIBE Books;
-- Step 5: Describe Members Table
DESCRIBE Members;
-- Step 6: Insert Records into Books Table
INSERT INTO Books VALUES
(1,'The Alchemist','9780061122415',1988),
(2,'Clean Code','9780132350884',2008),
(3,'Atomic Habits','9780735211292',2018);
```

Output

Action Output

#	Time	Action	Message
2	19:08:15	USE library_system_db	0 row(s) affected
3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHA...	0 row(s) affected
4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
5	19:10:02	DESCRIBE Books	4 row(s) returned
6	19:10:39	DESCRIBE Members	3 row(s) returned
7	19:11:13	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Week-1*

```
DESCRIBE Members;
-- Step 6: Insert Records into Books Table
INSERT INTO Books VALUES
(1,'The Alchemist','9780061122415',1988),
(2,'Clean Code','9780132350884',2008),
(3,'Atomic Habits','9780735211292',2018);
-- Step 7: Display Books Table
SELECT * FROM Books;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018
NULL	NULL	NULL	NULL

Books 3

Output

Action Output

#	Time	Action	Message
3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHA...	0 row(s) affected
4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
5	19:10:02	DESCRIBE Books	4 row(s) returned
6	19:10:39	DESCRIBE Members	3 row(s) returned
7	19:11:13	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	19:11:50	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

Week-1

```
    member_id INT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE
);
-- Step 4: Describe Books Table
DESCRIBE Books;
-- Step 5: Describe Members Table
DESCRIBE Members;
-- Step 6: Insert Records into Books Table
INSERT INTO Books VALUES
(1,'The Alchemist','9780061122415',1988),
(2,'Clean Code','9780132350884',2008),
(3,'Atomic Habits','9780735211292',2018);
-- Step 7: Display Books Table
SELECT * FROM Books;
-- Step 8: Insert Records into Members Table
INSERT INTO Members VALUES
(101,'Harshika','iliharshika@gmail.com'),
(102,'Bhanu Priya','bhanupriya@gmail.com'),
(103,'Nikshitha','nikshitha@gmail.com');
```

Output

Action Output

#	Time	Action	Message
4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
5	19:10:02	DESCRIBE Books	4 row(s) returned
6	19:10:39	DESCRIBE Members	3 row(s) returned
7	19:11:13	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	19:11:50	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
9	19:12:58	INSERT INTO Members VALUES (101,'Harshika','iliharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Week-1

```
SELECT * FROM Books;
-- Step 8: Insert Records into Members Table
INSERT INTO Members VALUES
(101,'Harshika','iliharshika@gmail.com'),
(102,'Bhanu Priya','bhanupriya@gmail.com'),
(103,'Nikshitha','nikshitha@gmail.com');
-- Step 9: Display Members Table
SELECT * FROM Members;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

member_id	full_name	email
101	Harshika	iliharshika@gmail.com
102	Bhanu Priya	bhanupriya@gmail.com
103	Nikshitha	nikshitha@gmail.com
NULL	NULL	NULL

Members 4

Output

Action Output

#	Time	Action	Message
5	19:10:02	DESCRIBE Books	4 row(s) returned
6	19:10:39	DESCRIBE Members	3 row(s) returned
7	19:11:13	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	19:11:50	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
9	19:12:58	INSERT INTO Members VALUES (101,'Harshika','iliharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	19:13:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned

Week-1

Limit to 1000 rows

```
18  DESCRIBE Books;
19  -- Step 5: Describe Members Table
20  DESCRIBE Members;
21  -- Step 6: Insert Records into Books Table
22  INSERT INTO Books VALUES
23  (1,'The Alchemist','9780061122415',1988),
24  (2,'Clean Code','9780132350884',2008),
25  (3,'Atomic Habits','9780735211292',2018);
26  -- Step 7: Display Books Table
27  SELECT * FROM Books;
28  -- Step 8: Insert Records into Members Table
29  INSERT INTO Members VALUES
30  (101,'Harshika','iliharshika@gmail.com'),
31  (102,'Bhanu Priya','bhanupriya@gmail.com'),
32  (103,'Nikshitha','nikshitha@gmail.com');
33  -- Step 9: Display Members Table
34  SELECT * FROM Members;
35  -- Step 10: Test Duplicate ISBN
36  INSERT INTO Books VALUES
37  (4,'Fake Book','9780061122415',2000);
```

Output

Action Output

#	Time	Action	Message
6	19:10:39	DESCRIBE Members	3 row(s) returned
7	19:11:13	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	19:11:50	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
9	19:12:58	INSERT INTO Members VALUES (101,'Harshika','iliharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	19:13:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
11	19:14:03	INSERT INTO Books VALUES (4,'Fake Book','9780061122415',2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'

Week-1

Limit to 1000 rows

```
24  (2,'Clean Code','9780132350884',2008),
25  (3,'Atomic Habits','9780735211292',2018);
26  -- Step 7: Display Books Table
27  SELECT * FROM Books;
28  -- Step 8: Insert Records into Members Table
29  INSERT INTO Members VALUES
30  (101,'Harshika','iliharshika@gmail.com'),
31  (102,'Bhanu Priya','bhanupriya@gmail.com'),
32  (103,'Nikshitha','nikshitha@gmail.com');
33  -- Step 9: Display Members Table
34  SELECT * FROM Members;
35  -- Step 10: Test Duplicate ISBN
36  INSERT INTO Books VALUES
37  (4,'Fake Book','9780061122415',2000);
38  -- Step 11: Test NULL Title
39  INSERT INTO Books VALUES
40  (5,NULL,'9999999999999',2010);
41  -- Step 12: Test Future Publication Year
42  INSERT INTO Books VALUES
43  (6,'Future Book','8888888888888',2030);
```

Output

Action Output

#	Time	Action	Message
7	19:11:13	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	19:11:50	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
9	19:12:58	INSERT INTO Members VALUES (101,'Harshika','iliharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	19:13:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
11	19:14:03	INSERT INTO Books VALUES (4,'Fake Book','9780061122415',2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
12	19:21:04	INSERT INTO Books VALUES (6,'Future Book','8888888888888',2030)	Error Code: 3819. Check constraint 'books_chk_1' is violated.

Week-1

Limit to 1000 rows

```
27  SELECT * FROM Books;
28  -- Step 8: Insert Records into Members Table
29  INSERT INTO Members VALUES
30  (101,'Harshika','iliharshika@gmail.com'),
31  (102,'Bhanu Priya','bhanupriya@gmail.com'),
32  (103,'Nikshitha','nikshitha@gmail.com');
33  -- Step 9: Display Members Table
34  SELECT * FROM Members;
35  -- Step 10: Test Duplicate ISBN
36  INSERT INTO Books VALUES
37  (4,'Fake Book','9780061122415',2000);
38  -- Step 11: Test NULL Title
39  INSERT INTO Books VALUES
40  (5,NULL,'9999999999999',2010);
41  -- Step 12: Test Future Publication Year
42  INSERT INTO Books VALUES
43  (6,'Future Book','8888888888888',2030);
44  -- Step 13: Test Duplicate Email
45  INSERT INTO Members VALUES
46  (104,'Harshika Clone','iliharshika@gmail.com');
```

Output

Action Output

#	Time	Action	Message
8	19:11:50	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
9	19:12:58	INSERT INTO Members VALUES (101,'Harshika','iliharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	19:13:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
11	19:14:03	INSERT INTO Books VALUES (4,'Fake Book','9780061122415',2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
12	19:21:04	INSERT INTO Books VALUES (6,'Future Book','8888888888888',2030)	Error Code: 3819. Check constraint 'books_chk_1' is violated.
13	19:21:59	INSERT INTO Members VALUES (104,'Harshika Clone','iliharshika@gmail.com')	Error Code: 1062. Duplicate entry 'iliharshika@gmail.com' for key 'members.email'